

ARNUR JUMABEKOV

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EDUCATION

Haileybury Almaty

High School Diploma (A-Levels)

Almaty, Kazakhstan

Aug 2023 – Jun 2025

- **A-Levels:** Computer Science (A*), Maths (A*), Further Maths (A*), Physics (A)
- **IELTS:** 8.0/9.0 | **SAT:** 1550/1600 (750 EBRW, 800 Math) | **AP Courses:** Calculus BC - 5/5, Computer Science A - 5/5

CERTIFICATIONS

Samsung Innovation Campus – Android Development Track | *Java, Google Firebase SDK* (2020 – 2021)

Microsoft AI Challenge – Machine Learning Track | *Python, PyTorch, scikit-learn, pandas, NumPy, matplotlib* (2023)

PUBLICATIONS

Arnur Jumabekov, Maryam Yashtini, “WeCAViT: A Weighted CNN model for Pneumonia Detection in Chest X-rays,” in Proceedings of the 13th IEEE International Conference on Intelligent Control and Information Processing (ICICIP 2025), March 2025. <https://ieeexplore.ieee.org/document/10898141>

PORTFOLIO

STEM Portfolio — Robotics & Other Projects (Slides)

EXPERIENCE

Smart Pipe Inspection Robot – Research Assistant

ARMS Lab, Nazarbayev University

Aug 2025 – Present

Astana, Kazakhstan

- Developing **Python** API to control Dynamixel DC motors; Implemented custom path-following algorithms

Tensegrity Robot Project – Intern, Lead Programmer

ISSAI, Nazarbayev University

May 2024 – Jun 2024

Astana, Kazakhstan

- Wrote C++ control firmware for **Arduino** boards; Integrated modular OOP architecture, streamlined mechanisms and ports' interfaces; Implemented **PID control** for linear actuators; fine-tuned for < 5% position error
- Developed **Python** pipelines for real-time position extraction via Serial Port for OptiTrack feedback matching

Pioneer Research Institute – Student Researcher

Supervised by Dr. Maryam Yashinti (Georgetown University, D.C., USA)

Jun 2024 – Jan 2025

Remote

- Conducted literature review on Computer Vision applications in Healthcare; Proposed and experimented on the WeCAViT architecture (see Projects)

AWARDS

International Science & Engineering Fair (ISEF) – Special Award (2025) | Awarded by Regeneron & "Mawhiba" Foundation

Republican Science Projects Fair (KazSEF) – Gold Medal (2025) | Top 0.7% of all projects, "Informatics" section

British Physics Olympiad – Silver Medal (2024) | Top ≈10% of all participants worldwide

PROJECTS

WeCAViT | *Python, PyTorch, Pandas, Scikit-Learn, Google Colab, Huggingface Transformers*

Jun 2024 – Present

- Designed a hybrid CNN-CBAM-ViT model for pneumonia detection; Achieved ~4-5% accuracy & F1 increase over several state-of-the-art CNNs, Vision Transformers (ViT), and hybrid models
- Conducted >17 ablation experiments, >8 comparisons with SOTA models
- Authored a research paper proposing WeCAViT model

QuantaSoil | *Python, Pandas, Scikit-Learn, Google Colab, OpenAI API*

Sep 2023 – Nov 2023

- Trained and evaluated ML classifiers to predict optimal crop/fertilizer based on soil data from open-source datasets; achieved > 95% accuracy
- Engineered custom text prompts for user interaction using GPT4 via OpenAI API; integrated in React web app

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, C++, (SQL)

Frameworks & SDKs: PyTorch, Google Firebase SDK

Tools: Jupyter, Git, Google Colab, VS Code, Android Studio, LaTeX

Libraries: pandas, NumPy, Matplotlib, Scikit-learn, Huggingface Transformers